

❖ FitTrack AI - Personalized fitness recommendations powered by AI.

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Description

The **FitTrack AI** backend is a secure, scalable, and AI-powered RESTful application developed to simplify workout tracking and provide intelligent fitness recommendations. Built using **Node.js**, **Express.js**, **MongoDB**, and **Mongoose**, the backend exposes REST APIs that enable secure user authentication, workout management, workout searching, and AI-assisted fitness guidance.

The application follows the **Model-View-Controller (MVC)** architecture, making the project modular, maintainable, and scalable. User authentication is implemented using **JSON Web Tokens (JWT)**, while passwords are securely encrypted using **bcrypt.js** before storage in MongoDB. Every protected API validates the JWT token through middleware before allowing access to sensitive resources.

One of the key features of the project is its integration with **Google Gemini AI**, which provides personalized workout recommendations based on a user's age, fitness goals, and workout experience. The AI module also generates fitness insights by analyzing workout statistics such as total workouts completed, average workout duration, and calories burned.

BRANSTORMING & IDEATION PHASE

Scenario-Based Case Study

Background

Rahul is a fitness enthusiast who exercises regularly but records his workouts manually using notebooks and spreadsheets. Although he maintains consistency in his fitness routine, he finds it difficult to organize workout history, monitor progress, and determine whether his exercise habits align with his fitness goals. He also lacks professional guidance when deciding which workouts are best suited for his current fitness level.

PROBLEM

- Difficulty maintaining a centralized record of workouts.
- Manual tracking leads to inaccurate workout history.
- No secure user authentication for personal fitness records.
- Unable to search previous workouts efficiently.
- No personalized workout recommendations.
- No insights regarding overall workout performance.
- Time-consuming manual calculation of calories burned and workout duration.

Solution

The **FitTrack AI** provides a centralized backend system where users can securely register, authenticate, and manage workout records. Users can add, update, delete, retrieve, and search workouts through RESTful APIs. By integrating **Google Gemini AI**, the application analyzes user information and workout statistics to generate intelligent workout recommendations and personalized fitness insights.

The backend follows industry-standard security practices including JWT authentication, password hashing, middleware-based authorization, centralized error handling, and MongoDB schema validation.

Usage

User Usage

A registered user can create an account using the registration endpoint and securely log in using email and password. After successful authentication, the backend generates a JWT token that authorizes access to protected APIs.

Using authenticated APIs, users can:

- Create new workout records.
- View all previously completed workouts.
- Retrieve a workout by its unique identifier.
- Update existing workout information.
- Delete unwanted workout records.
- Search workouts using workout name, category, or workout date.
- Generate AI-powered workout recommendations.
- Obtain AI-generated fitness insights based on workout statistics.

AI Usage

The AI module communicates with Google Gemini using a dedicated service layer.

Users provide information such as:

- Age
- Fitness Goal
- Experience Level

Gemini processes this information and returns:

- Personalized workout plan
- Weekly exercise suggestions
- Beginner/Intermediate/Advanced recommendations
- Motivational guidance
- Safety tips

Similarly, users can submit workout statistics including:

- Total Workouts
- Average Workout Duration
- Total Calories Burned

The AI module analyzes these values and produces personalized fitness insights.

Outcome

The AI FitTrack backend significantly improves fitness tracking by eliminating manual record keeping and providing intelligent recommendations based on user activity. Secure authentication ensures that only authorized users can access personal workout data, while AI assistance helps users improve their exercise routines with personalized suggestions.

The modular MVC architecture enables easy maintenance and future scalability. The application can later be extended with additional features such as nutrition tracking, BMI calculation, wearable device integration, workout reminders, and fitness analytics dashboards without major architectural changes.